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IS 475 – 1001

11/30/2021

Automobile Service and Inventory System

Background and business problems: Identify and define the business problem to be solved in an organization.

- While my project idea will primarily focus on the automotive repair sector or service stations throughout the world, many businesses face the same challenge. That is, the ability to keep track of, manage, and maintain a company's inventory. Companies must identify and know how much inventory they presently have to avoid shortages of high-demand products.

Furthermore, firms must attempt to recognize any patterns in inventory stock and adjust when the trend shifts. This is one of the reasons why businesses hire people who are aware of and can use technology that can assist them in assessing the data they see. Companies may benefit themselves as well as their consumers by using good inventory management. Knowing the precise amount of stock in their inventory allows them to understand which items are in more demand and which are not. Knowing this, firms may order more items in great need and less of what isn't selling well or has a low demand. Ultimately, a business can save a lot of money and at the same time be able to improve its cash flow. Customer satisfaction will also benefit from effective inventory management since they will receive the required items without waiting for them to become available. These are just some of why proper utilization and implementation of a database is essential to run a business successfully and adequately.

Data requirements: Summarize the business rules for the database. The requirements should describe the relationships between data components in the business. Describe the data that will be included in this project, including structure data and possibly unstructured data.

- The data needed for my planned vehicle repair and inventory control database will begin with the customer. Because an auto repair service is primarily dependent on clients for business, each customer must be recognized by a unique customer ID for my proposed database. This enables the database to link the customer to their scheduled appointments, service bill, vehicle, and mechanic in charge of their car. They'll also provide their first name, last name, as well as their phone number. As previously stated, each customer must plan an appointment to avoid scheduling conflicts. Therefore, customers will be provided with a unique appointment number, a date, and the time for their vehicle service. In addition, customers may bring many automobiles to the business; thus, the database must record the unique VIN of each vehicle and the manufacturer, model, and year of production. A unique service ID is granted when a vehicle enters service to minimize confusion and record the service date. A vehicle will also have only one mechanic assigned to service a customers vehicle. The database will keep track of the unique mechanic's ID, first name, last name, and phone number. The proposed database will also allow the user to view the component used by the mechanic, the cost of that part, and the type of part it is. This will provide the business data they can utilize to have a solid and successful business strategy.

Business value: Articulate the business value of the database system to be developed.

- My suggested database system is primarily focused on preventing and reducing any disputes or confusion, not only with consumers but also with the business's profitability. Confusion within a corporation may have a detrimental impact on both the organization and its consumers. For example, if a mechanic services a car with the same make, model, and color but not the VIN allocated to them, this might result in monetary losses. In addition, consumers will be extremely upset and dissatisfied due to the company's blunder, which can deteriorate a company's reputation and cause loss of customers. Not only that, but the database will allow car repair firms to view the components used by technicians and the quantity utilized, the type of item used, the amount required to finish the job, and how much the parts cost the business to acquire. This will help firms discover a pattern that might save them money in the future by working with firms that specialize in specific components or reducing the number of parts bought that are not typically utilized in the business.